

CARGOMULE · REQUIREMENTS

CargoMule requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cargomule		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R8 relaxed to 45 kg; R3 restated with thermal derating and a use limit; status from CGM-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3. R8 relaxed to 40 kg and R12 redefined as the \$1,000 budget (CGM-DDR-001); status from CGM-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

CargoMule requirements

These requirements are checked by calculation in CGM-CAL-001 v0.2 at TRL 3. Targets changed with Amish's decisions of 2026-09-25. Under CGM-DDR-001, R8 was relaxed from 35 kg to 40 kg for the steel frame (D3) and R12 was redefined as the \$1,000 prototype budget (D1). Under CGM-DDR-002, R8 is relaxed again to 45 kg for the first prototype, and R3 is restated: the 300 m climb target stands, and beyond it the controller derates motor current to protect the winding, with the use limit stated to riders. Against the calculations, **none is missed and three are at risk (R3, R6, R7)**; nine are met on paper or by design, and two cannot be verified at TRL 3. Targets are not yet validated with users; the first users and region for co-design are still open (CGM-DDR-001, O1).

The **design load case** is 150 kg of payload centered on the deck, the trailer (about 43 kg, CGM-CAL-001) and a towing bicycle with rider of about 100 kg, on a dry paved road. The **design route** is a 10 km round trip with 120 m of climbing, 120 m of descent and 20 stops.

Table 1. Requirements, targets and status at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CGM-CAL-001)
R1	Carry the payload	150 kg (330 lb) on a deck of at least 0.8 m ² , with tie-down points	Frame stress calculation; later proof load	Met on paper: deck 0.84 m ² ; least structural factor 1.6 at 2 g and the ultimate hitch case
R2	Range per charge	20 km or more on the design route in the design load case	Energy calculation; later logged ride	Met on paper: 24.9 km
R3	Assist on hills	8 % grade for 300 m at 8 km/h or more with the rider feeling 40 N or less of drawbar pull, without motor over-temperature at 30 °C. On longer continuous climbs the controller derates motor current so the winding stays at or below 100 °C; the use limit (about 300 m of continuous 8 % climbing at full assist) is stated in the rider instructions (restated, CGM-DDR-002)	Force and thermal calculation; motor datasheet at part selection	At risk: 36.6 N felt; winding about 95 °C against a 100 °C limit on assumed motor constants; derating starts after about 336 m [C10]
R4	Near-zero added load on the flat	Steady drawbar pull felt by the rider 15 N or less at 5 to 20 km/h	Control calculation and simulation	Met on paper: 6.2 to 6.6 N; loop stable for G = 2 to 6 with the 0.7 Hz filter
R5	Stay within pedal-assist limits	250 W rated continuous motor output; assist only while the drawbar is in tension; no assist above 25 km/h; no throttle	Datasheets; controller settings	Met by design
R6	Trailer brakes itself	With the combination slowing at 3 m/s ² , the trailer pushes the bicycle with 100 N or less; assist cut within 100 ms of drawbar compression above 20 N	Braking calculation; later bench test	At risk: 92 N dry and about 110 N with wet pads on 180 mm rotors with metallic pads (CGM-DDR-002); rotor heating on long descents; cut about 60 ms on paper
R7	Hitch to common bicycles	Fits rear wheels with a 9 mm quick release or a 12 mm thru-axle, 130 to 148 mm spacing, 26 in to 700c; hitch and unhitch in 30 s or less; no wiring to the bike	Hitch design review; later fit survey	At risk: axle stresses pass; thru-axles need a hitch axle per thread pitch and length; nutted axles, some hub gears and some rear-motor e-bikes need adapters
R8	Light enough to handle	Empty trailer 45 kg (99 lb) or less, including the pack, for the first prototype (relaxed from 35 kg to 40 kg by CGM-DDR-001 D3, then to 45 kg by CGM-DDR-002)	Mass estimate; later weighing	Met on paper: 43.2 kg (40.4 kg without side boards)
R9	Fit bike paths and doors	Overall width 1,000 mm or less; length 2.6 m or less hitched from the bike's rear axle	Model check	Met on paper: 960 mm wide, 2.52 m long
R10	Stable hitch load	Downward load on the hitch 3 to 10 kg with the design payload centered	Mass and balance estimate	Met on paper: 7.8 kg; holds only for a payload

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CGM-CAL-001)
				center 28 mm ahead to 62 mm behind the deck center
R11	Weather and temperature	Electronics IP65; motor IP54 or better; operate –10 to 40 °C; the BMS blocks charging below 0 °C	Datasheets and design review	Not verifiable at TRL 3
R12	Affordable	Prototype parts \$1,000 or less, including charger, equal to <code>budget_usd</code> (redefined, CGM-DDR-001 D1)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Met on paper: \$979
R13	Fail safe	Loss of the load cell signal, a reading out of range, a stuck value or a controller fault sets motor current to zero within 100 ms; the trailer stays towable as an unpowered, braked trailer	Fault tree and design review	Not verifiable at TRL 3 (60 ms fast path on paper)
R14	Be seen	Rear red lights and reflectors, side reflectors, and a flag at 1.5 m or more above the ground	Design review	Met by design: flag top 1,570 mm

■ Assumptions

- Rolling resistance coefficient 0.010 for 20 x 2.15 in tyres on pavement at design load; extra drag area of the loaded trailer 0.1 m², since it rides partly in the rider's wake.
- Air density 1.2 kg/m³; gravitational acceleration 9.81 m/s².
- Drive efficiency from a motor model at each operating point (CGM-CAL-001, section C) rather than a fixed 75 %; charger efficiency 90 %; cell charging efficiency 96 % for LiFePO₄.
- Usable pack energy 90 % of the nominal 384 Wh.
- 250 W and 25 km/h follow the EU pedal-assist exclusion (Regulation (EU) No 168/2013); the US federal definition allows more (under 750 W, 20 mph), so the EU limits are the tighter case. Whether a motorized trailer falls under either rule is open (CGM-DDR-001, O2).
- Brake and hitch loads follow the voluntary cycle trailer standards EN 15918 and ASTM F1975, which limit unbraked trailers to 60 kg and 45.4 kg; CargoMule exceeds both, so its brakes are required, not optional.